

# ECE 315: Signals and Systems

## Homework 3

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This homework is due on **2/20, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (1pt) Given a discrete-time system's impulse response  $h[n]$ , we can compute the system's output for any input signal  $x[n]$  through convolution sum. We consider the following infinite-horizon cases:

- $h[n] = u[n]$  and  $x[n] = u[n]$
- $h[n] = 2^n u[n]$  and  $x[n] = u[n]$
- $h[n] = 2^n u[n]$  and  $x[n] = nu[n]$
- $h[n] = nu[n]$  and  $x[n] = nu[n]$

Compute the system's output for each case described above.

2. (1pt) Given a continuous-time system's impulse response  $h(t)$ , we can compute the system's output for any input signal  $x(t)$  through convolution integral. We consider the following infinite-horizon cases:

- $h(t) = e^t u(t)$  and  $x(t) = u(t)$
- $h(t) = e^t u(t)$  and  $x(t) = e^t u(t)$
- $h(t) = tu(t)$  and  $x(t) = t^2 u(t)$
- $h(t) = tu(t)$  and  $x(t) = e^{2t} u(t)$

Compute the system's output for each case described above.

3. (2pt) Given a discrete-time system's impulse response  $h(t)$ , we can compute the system's output for any input signal  $x(t)$  through convolution sum. We consider the following finite-horizon cases:

- $h[n] = u[n - 1] - u[n + 2]$  and  $x[n] = 2^n u[-n]$
- $h[n] = u[n] - u[n + 2]$  and  $x[n] = 2^n (u[n - 1] - u[n + 4])$

Compute the system's output for each case described above.

- (2pt) Given a continuous-time system's impulse response  $h(t)$ , we can compute the system's output for any input signal  $x(t)$  through convolution integral. We consider the following finite-horizon cases:

- $h(t) = \begin{cases} t^2 & \text{when } 0 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$  and  $x(t) = u(t - 1) - u(t + 1)$
- $h(t) = \begin{cases} 2^{-t} & \text{when } 0 \leq t \leq 4 \\ -2^t & \text{when } -2 \leq t < 0 \end{cases}$  and  $x(t) = u(t)$

Compute the system's output for each case described above.

- (2pts) Prove that for any linear time-invariant (LTI) system, its impulse response is uniquely determined. *Hint:* Prove it by contradiction. Assume there are two different impulse responses and argue this is impossible for a linear time-invariant system. Moreover, come up with an example that shows two different non-LTI systems can have the same impulse response.
- (2pts) Consider the following cascaded system (w/ input  $x[n]$  and output  $y[n]$ ) that consists of two subsystems:
  - System 1 with impulse response  $h_1[n] = u[n - 2]$ .
  - System 2 with impulse response  $h_2[n] = 2^n u[-n]$ .



If we take an input  $x[n] = \delta[n] - \delta[n - 4]$ , we next determine the output  $u[n]$  in two different ways.

- Compute the output  $y_1[n]$  of System 1. Using  $y_1[n]$  as an input of System 2, compute the output  $y[n]$  of System 2.
  - Compute the convolution sum of two impulse responses  $h[n] = h_1[n] * h_2[n]$ . Viewing  $h[n]$  as an impulse response of a system, compute the output when input is  $x[n]$ . Is this output different from the one in (i)?
- Read Sections 3.8-1, 3.11-1, and 3.11-4, and complete the MATLAB problem 3.11-3 in LSS.